PlusMinus Analytics:

What "I" Did

- Developed a basketball analytics prototype and the product documentation around it.
- Organized the concept around 3 capabilities:
  - Understanding shooting results.
  - Reviewing session performance.
  - Turning those results into a practice plan.
- Connected data definitions, transformations, validation, and documentation to a specific user decision.
- Interview theme: I worked on making data trustworthy, understandable, and useful.

How I introduce the project

- PlusMinus has 2 separate experiences: a historical NBA explorer and a training-product prototype.
- The historical component uses 79,763 exported shots across 10 selected players, with observed game dates spanning 1996-2026.
- Python transformations clean shot records, calculate shooting metrics, and aggregate results for visual analysis.
- Training Lab uses synthetic data and explicit rules to demonstrate a practice-planning workflow.
- The strongest work to discuss: correcting era classification, defining metrics, testing relationships and calculations, and documenting a focused next release.
- A data product needs clear definitions and reliable transformations in addition to a dashboard.

Why this relates to Vanguard

- Target role: Data Product Analyst, Personal Wealth Data Strategy & Management.
- Data discovery: understand the grain, source, coverage, and intended use of a dataset.
- Data quality: identify problems, trace their causes, and validate the correction.
- Requirements: document business rules, definitions, acceptance criteria, and success measures.
- Communication: explain decisions and limitations clearly to product, engineering, and business partners.
- Use the project as evidence of relevant preparation; do not claim enterprise, cloud, or cross-functional delivery experience that did not occur.

How to use these notes

- These notes consolidate the substantive interview and resume discussion from this chat in the reference PDF's note style.
- First-person wording is a rehearsal draft. Use built, reviewed, helped develop, or validated according to my actual contribution.
- The project used AI coding assistance. Be ready to explain the decisions and validation I personally owned.
- STAR = Situation, Task, Action, Result. Practice stories 1, 3, and 6 first.

STAR 1: Correcting a data-quality issue

When I use this answer

- Tell me about a difficult problem, your attention to detail, or how you validate data.

Situation

- The NBA explorer grouped shots into eras to compare shooting patterns over time.
- The existing transformation could put a home-team abbreviation into the era field under some conditions.
- It also mishandled compact season identifiers such as 22023.
- A populated field and plausible chart could therefore communicate the wrong meaning.

Task

- Make era classification represent the correct basketball season and carry the correction through exported data and dashboard labels.

Action

- Traced the era field back to the transformation that created it.
- Defined source precedence: use explicit season metadata first; infer from game date only when needed.
- Normalized supported season formats so 22023 represents the season beginning in 2023.
- Accounted for calendar year versus season year. January games generally belong to a season that started the previous year.
- Included a specific exception for the 2020 bubble when using date inference.
- Updated era values in the exported shot records while preserving other fields and row order.
- Rebuilt era-level aggregates in both export locations.
- Added regression cases for metadata precedence, compact identifiers, invalid dates, mixed formats, and frontend label coverage.

Result

- The corrected implementation produces season-based era classifications and includes coverage for known failure cases.
- The grouping now matches the meaning intended by the dashboard comparison.
- Lesson: structurally valid data can still be semantically wrong.
- Describe the discovery as my own only if I personally found it; otherwise explain how I reviewed, understood, or validated the repair.

Concepts I can explain

- Semantic quality: a team abbreviation is valid text, but not a valid era.
- Source precedence: authoritative metadata takes priority over a date approximation.
- Lineage: follow a value from source records through transformations, exports, and charts.
- Regression testing: turn known failures into repeatable checks.
- Vanguard connection: business decisions depend on fields having consistent, correct meaning.

STAR 2: Understanding the data

When I use this answer

- How do you approach a new dataset? What data discovery or profiling have you done?

Situation

- The project contained historical NBA exports and a separate training dataset.
- Both involved basketball shooting, but their origins, structures, and intended uses differed.
- Without those distinctions, the analysis could imply unsupported comparisons or representativeness.

Task

- Clarify what each dataset represents, how records relate, and which questions each source can reasonably answer.

Action

- Documented the grain: one historical shot record represents one shot attempt.
- Documented training relationships: events connect to sessions; sessions connect to players.
- Recorded historical coverage: 79,763 shots, 10 selected players, observed game dates from November 1, 1996 to February 23, 2026.
- Treated this as selected-player sample coverage, not a complete league dataset or proof of collection freshness.
- Documented synthetic training data: 3 fictional players, 24 sessions, and 1,440 shot events.
- Kept historical NBA records separate from fictional practice records; they are not joined to establish player performance.
- Recorded assumptions: synthetic sessions use equal shot volume; historical era comparisons lack adjustments for pace, opponents, and rule changes.

Result

- The documentation makes source, coverage, relationships, and limitations explicit.
- That supports choosing an appropriate analysis and explaining what a result can support.
- Lesson: establish what the data represents before looking for an interesting pattern.

Concepts I can explain

- Grain: the unit represented by one record. Mixing grains can duplicate records or distort aggregates.
- Provenance: where data came from and how it was generated.
- Coverage versus completeness: a wide date range does not prove complete history.
- Representativeness: a selected sample does not automatically support population-wide conclusions.
- Synthetic data: useful for reproducible demonstrations and testing; not evidence of actual user behavior.
- Vanguard connection: understand the dataset and business context before recommending how to use it.

STAR 3: Defining and validating metrics

When I use this answer

- How do you ensure reporting is accurate? Explain a technical concept to a business stakeholder.

Situation

- Training Lab shows accuracy across players, sessions, and zones.
- Different aggregation methods can produce different answers.
- There was also a risk of describing a shooting percentage as a broader measure of skill.

Task

- Define metrics consistently and connect displayed values back to individual shot events.

Action

- Defined overall accuracy as total makes divided by total attempts.
- Weighted each attempt equally rather than taking an unweighted average of session percentages.
- Kept an empty denominator unknown: no attempts is different from missing every shot.
- Distinguished percentage-point change from relative percentage change.
- Used tests to reconcile zone attempt totals to player events.
- Recalculated stored session ratings from each session's shots.
- Documented the shooting index as session shooting percentage on a 0-100 scale, not an independently validated skill score.

Result

- The calculations have explicit definitions and automated reconciliation checks.
- The dashboard is easier to explain because the same measure has a consistent meaning.
- Lesson: explain the numerator, denominator, population, and time period before interpreting a number.

Example I can say out loud

- Session 1: 1 make out of 2 attempts = 50%.
- Session 2: 9 makes out of 10 attempts = 90%.
- A simple average gives 70%, but total makes / total attempts = 10 / 12 = 83.3%.
- For overall shot accuracy, I use total makes divided by total attempts.
- A move from 40% to 50% is 10 percentage points, or a 25% relative increase.

Concepts I can explain

- Reconciliation: check that a summary agrees with the records underneath it.
- Missing versus zero: no evidence and poor performance are different states.
- Metric definition: specify calculation, scope, unit, and treatment of missing data.
- Vanguard connection: shared definitions help teams use the same metric consistently.

STAR 4: Modeling related records

When I use this answer

- What SQL or data-modeling experience do you have? How do you validate related datasets?

Situation

- The training experience connects players, sessions, shot maps, events, ratings, and plans.
- Records can reference existing IDs and still connect to the wrong player or session.

Task

- Make relationships explicit and validate both references and ownership consistency.

Action

- Worked with 7 entities and documented primary keys and relationships.
- Separated referential integrity from relationship consistency.
  - An event's shot map must exist and belong to the same session as the event.
  - A rating's player must be the player who owns the referenced session.
- Used composite foreign keys in the SQL schema to express paired relationships.
- Validated unique IDs, foreign-key references, event/session consistency, and rating ownership in fixture tests.
- Documented 5 simulated API contracts with required inputs, success responses, and error behavior.
- Kept architecture claims precise: browser JSON and an in-memory dispatcher power the prototype; SQLite is an inspectable artifact, not its runtime backend.

Result

- The model and contracts can be inspected and tested independently of the interface.
- The Node suite passed its 4 tests during this chat: relationships, metrics, plans, and route behavior.
- Lesson: reliability includes how records connect, not just whether individual values look reasonable.

Concepts I can explain

- Primary key: uniquely identifies a record.
- Foreign key: requires a reference to an existing record.
- Composite foreign key: validates a combination, such as session and player together.
- Data contract: defines expected inputs, outputs, meanings, and failure behavior.
- Relationship consistency is not access authorization. A production backend still needs authentication and server-side ownership checks.
- Vanguard connection: useful preparation for documenting rules and supporting validation with engineers.

STAR 5: Explaining a recommendation

When I use this answer

- Tell me about a tradeoff. How do you turn analysis into action or evaluate an AI-enabled idea?

Situation

- Training Lab needed to connect shot results to a practice plan users could understand and inspect.

Task

- Demonstrate a transparent recommendation process with testable rules and clearly stated limits.

Action

- Used a deterministic rule rather than a learned model.
- Calculated accuracy by zone for the selected player.
- Excluded zones with fewer than 20 attempts and selected the lowest-accuracy eligible zone.
- Used a consistent alphabetical tie-breaker when accuracies match.
- Included the selected zone, accuracy, and attempt count in the explanation.
- Allocated approximately 20% of duration to warm-up, 60% to focus, and the remainder to review.
- Handled integer rounding so allocated minutes sum exactly to the requested duration.
- Tested focus selection and durations across the fixture's players, plus invalid durations and missing evidence.
- Documented the limits: raw low accuracy may reflect a harder shot type, and 20 attempts is a heuristic rather than statistical confidence.

Result

- The prototype demonstrates a traceable path from records to an action.
- The limitation informed the proposed next release: let the player choose a focus and present the relevant evidence.
- It does not demonstrate technique diagnosis or measured player improvement.

Example I can demonstrate

- The documented fictional Jordan Ellis demo has 54.6% overall accuracy.
- Above-break 3 is the selected focus: 40.0% across 120 attempts.
- A 45-minute plan allocates 9 minutes to warm-up, 27 to focus, and 9 to review.
- I can explain the evidence, rule, and allocation rather than treating the output as a black box.

Concepts I can explain

- Deterministic: identical inputs produce the same result.
- Explainable: the output exposes the records and logic behind it.
- Personalized here means tailored to recorded results and chosen time, not learned understanding of technique.
- Vanguard connection: assess whether an analytical output is explainable and appropriate for its decision; this is not a financial recommendation model.

STAR 6: Defining the next release

When I use this answer

- How do you prioritize, document requirements, or define success?

Situation

- The broad vision covered historical analysis, player and coach dashboards, plans, and possible AI features.
- The product question was which single workflow a real user should be able to complete.

Task

- Specify a focused next release with observable acceptance criteria and a way to evaluate usefulness.

Action

- Narrowed the proposed MVP to an adult recreational player logging practice, reviewing results, choosing a plan, and recording a comparable follow-up.
- Translated the workflow into acceptance criteria:
  - Saved data survives refresh; retrying a save does not create duplicates.
  - One user cannot access another user's records.
  - Corrections update aggregates and mark affected plan evidence outdated.
- Matched the model to actual input: store manually entered zone totals without inventing shot locations.
- Specified evidence snapshots and rules versions so old plans retain their original rationale.
- Defined proposed usability checks and pilot measures; deferred complexity until the core workflow can be evaluated.

Result

- The deliverable is a focused PRD with scope, business rules, acceptance criteria, and proposed validation gates.
- This is a planning result, not a launched application or completed user study.
- Its value is converting an idea into work that can be implemented and checked.

How I would measure success

- Proposed observed usability sessions: can at least 4 of 5 participants save a session, locate a plan, and explain the evidence unaided?
- Proposed logging measure: median entry time of 90 seconds or less for prepared four-zone totals.
- Proposed pilot: 5-10 adults over 4 weeks; measure saved plans followed by linked sessions within 7 days.
- These are initial decision rules, not achieved results or statistically conclusive findings.

Concepts I can explain

- Requirement = what must happen. Acceptance criterion = observable proof it happened.
- Idempotency = retries do not create unintended duplicates.
- Evidence versioning = preserve the information supporting an earlier decision.
- Vanguard connection: help translate business needs into testable delivery work.

Resume 1: Product scope and roadmap

Original bullet discussed

- Built an AI-powered basketball analytics MVP, producing the full PRD, feature blueprint, UX flows, and a 6-month roadmap covering shot tracking, skill ratings, and personalized training.

What I actually show

- A basketball analytics prototype and supporting product documentation.
- PRD means Product Requirements Document:
  - Describes the problem and intended user.
  - Defines requirements and acceptance criteria.
- Feature blueprint: breaks the experience into specific capabilities, inputs, behavior, and outputs.
- UX flows: show how a user moves from reviewing evidence to a next action.
- Six-month roadmap: proposed sequencing, dependencies, and validation milestones, not six months of completed delivery.

What makes it personalized?

- Uses the selected player's records and calculates accuracy by zone.
- Considers zones with at least 20 attempts and selects the lowest-accuracy eligible zone.
- Allocates the selected duration across warm-up, focus, and review.
- This is rules-based tailoring, not a model that learns a player's technique.

Why 6 months?

- Six months is the proposed planning horizon.
- It shows a development sequence beyond the initial prototype.
- It is an estimate, not a proven delivery timeline. Add a more specific rationale only if it reflects how I chose it.

Terms I should use precisely

- AI-powered: not supported by current runtime behavior; AI explanations are proposed.
- Shot tracking: synthetic shot records and visualizations, not automated video detection.
- Skill ratings: descriptive session shooting percentage, not validated skill measurement.
- MVP: current runnable prototype and proposed real-user next release are distinct.
- Static delivery supports an inspectable demo; it does not establish authenticated writes, durable sessions, or backend scale.

Revised resume wording

- Developed a basketball analytics prototype with rules-based practice recommendations; documented the PRD, feature blueprint, UX flows, and proposed six-month roadmap for session tracking, shooting metrics, and personalized practice plans.

Resume 2: How I got the numbers

Original bullet discussed

- Developed a 7-entity mock database enabling 15+ analytics visuals, 5 simulated API endpoints, and 3 interactive dashboards showcasing technical feasibility.

How I counted the entities

- Counted distinct entity types, not the number of records.
- Coaches: 1. Players: 3. Sessions: 24. Shot maps: 24.
- Events: 1,440. Ratings: 24. Initial example plans: 3.
- The browser uses 7 JSON collections. A corresponding SQLite schema and populated artifact make the model inspectable.
- Seven is the resulting model size, not a target. A simpler zone-total product may need a different model.

How I counted the visuals

- 3 dashboard views with 5 numbered visuals each = 15.
- Player: session accuracy, attempts by session, zone accuracy, shot map, shooting index.
- Coach: roster accuracy, practice volume, latest-versus-first change, time invested, player-by-zone matrix.
- Training: target gap, practice distribution, cumulative accuracy, made-versus-missed totals, practice allocation.
- Use 15 instead of 15+ unless additional visuals are explicitly inventoried in the same scope.
- These are not 15 independent measures: session accuracy and shooting index share the same underlying metric.

How I counted and tested the contracts

- GET /api/players; GET /api/sessions; GET /api/shot-maps; GET /api/ratings; POST /api/plans.
- Five method/path contracts execute inside the browser; they are not hosted HTTP endpoints.
- Tests cover all 5 contracts and missing IDs, unknown players, invalid duration, unsupported method, and unknown route.
- Generated plans are transient outputs; downloading a plan does not persist a database write.

What technical feasibility means

- Related records feed dashboard calculations and a traceable plan in a working browser experience.
- It does not establish production scale, secure multi-user operation, adoption, or recommendation effectiveness.
- These counts measure implementation scope. Tests check specific correctness properties.

Revised resume wording

- Modeled seven related entities in a synthetic dataset supporting 15 analytics visuals across three interactive dashboards and five simulated API contracts; validated relationships, metric reconciliation, and recommendation behavior with automated tests.

Resume 3: Case study and engagement

Original bullet discussed

- Produced a complete GitHub case study with architecture diagrams, competitive analysis, KPI framework, and GTM strategy, boosting project clarity and recruiter engagement by an estimated 3x.

What I can substantiate

- The case study connects the problem, architecture, implementation, decisions, evidence, and limits.
- Supporting documents cover competitive positioning, proposed KPIs, and go-to-market strategy.
- Separating implemented work from future plans helps a reviewer assess the project.
- Verify the files are committed and visible before describing the case study as published on GitHub. The earlier working-tree review showed untracked case-study files.

If asked: How did you measure 3x?

- That was an informal estimate, and I do not have a measured baseline supporting a threefold increase.
- The wording is too strong; I would remove it. I can substantiate the documentation, not the engagement uplift.
- An estimate still needs a definition, baseline, and method. Do not invent views, feedback, or response rates.

How I could measure engagement

- Define case-study engagement rate = unique visitors who open the case study / unique portfolio visitors.
- Illustrative example only: before, 10 of 100 visitors = 10%; after, 30 of 100 = 30%.
- 30% / 10% = 3x the rate; the change is 20 percentage points or a 200% relative increase.
- Use comparable windows, consistent tracking, and account for different traffic sources.
- A before/after comparison alone does not prove the case study caused the difference.
- Call it visitor engagement unless there is evidence the visitors are recruiters.

How I could measure clarity

- Ask reviewers to identify the problem, distinguish built versus proposed features, and explain a limitation.
- Record task completion and time to find supporting evidence.
- That evaluates communication; it does not automatically establish hiring impact.

Revised resume wording

- Created a portfolio case study documenting architecture, data provenance, validation results, competitive analysis, proposed KPIs, and go-to-market strategy, with clear distinctions between implemented capabilities and future work.

Follow-up answers and evidence

What did I personally do?

- I personally owned [the components I actually owned]. I used AI coding assistance for [the tasks it supported].
- My contribution was [specific decisions, analysis, implementation, review, or testing]. I can walk through [a function, rule, or test I understand].
- Do not claim independent discovery or authorship if I reviewed assisted work. Describe reviewing, testing, modifying, and explaining accurately.

How I explain the metrics to Vanguard

- I separate delivery measures from outcome measures.
- Seven entities, 15 visuals, 5 contracts, and 3 dashboards are verifiable implementation counts.
- Tests provide evidence about specific correctness properties.
- Adoption, better decisions, player improvement, and recruiter engagement require separate measurement.
- I want the evidence behind a claim to be as clear as the claim itself.

Boundaries to remember

- Historical data is a selected-player sample, not the entire NBA.
- Training Lab is synthetic and rules-based, not a trained AI product.
- The API is simulated; the prototype does not prove a live backend or production deployment.
- The next-release PRD, pilot, revenue, retention, and improvement outcomes are not completed achievements.
- Do not claim stakeholder interviews, Agile teamwork, cloud operations, or enterprise scale unless I actually did them.
- CSV export support does not prove a Tableau workbook was built; the reviewed repository had no committed workbook.

Where the evidence lives

- src/data_cleaning.py: cleaning, season classification, and metric transformations.
- scripts/repair_eras.py and tests/test_eras.py: export repair and era regressions.
- public/prototype/model.mjs: calculations, plan rules, and simulated contracts.
- tests/prototype.test.mjs: relationship, metric, plan, and contract checks.
- docs/recruiter/DATA_API.md and schema.sql: definitions, counts, and relationships.
- docs/recruiter/PRODUCT_SPEC.md, MVP_PRD.md, and STRATEGY.md: requirements and proposed strategy.
- docs/recruiter/CASE_STUDY.md, AUDIT.md, LAUNCH_KIT.md, and VERIFICATION.md: narrative, claims, demo, and recorded checks.

Verification context

- During this chat, all 4 Node tests passed. The earlier Python rerun could not start because the configured executable was unavailable.
- The existing October 10, 2026 report records prior Python, SQLite, and browser checks; those were not all rerun for these notes.
- Source: this chat and its repository review. Style reference: Plus Minus Evaluation.pdf. Prepared October 11, 2026.